AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2017

Physics Paper II

Time: 2 hours 20 minutes Marks: 55

AKU May Examination 2017

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are SIXTEEN questions. Answer ALL questions. Questions 14, 15 & 16 each offers TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

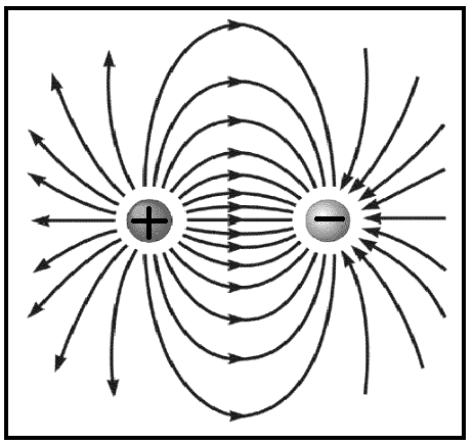
   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 3 Marks)

Name and describe the phenomenon illustrated in the given diagram.



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Q.2. (Total 4 Marks)

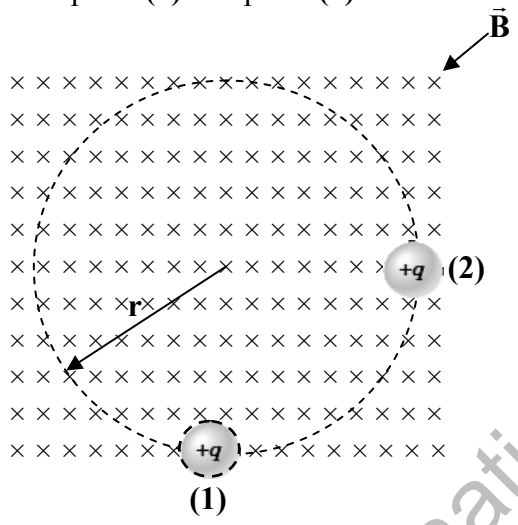
Differentiate between conventional and non-conventional current.

S.No.	Conventional Current	Non-Conventional Current
1		
2		
3		
4		

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Q.3. (Total 5 Marks)

i. Magnetic force helps to move a particle in a circular orbit. If a charged particle $+q$ is moving in a circular orbit with radius **r** in a magnetic field $\vec{B}$, then compare the particle's velocity $\vec{v}$ and the magnetic force $\vec{F}$ acting on it at point **(1)** and point **(2)**. (2 Marks)

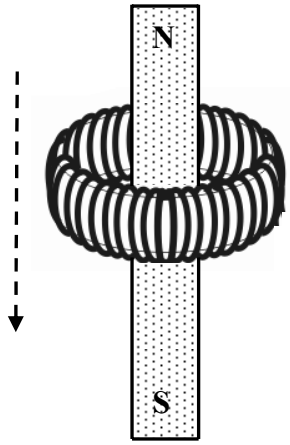


ii. Derive a mathematical relation to show that the radius of the given circle is inversely proportional to the magnitude of the magnetic field. (3 Marks)

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Q.4. (Total 2 Marks)

If a coil and a bar magnet are falling with the same velocity as shown in the given figure.



Will there be any induced current in the coil? Support your answer with a valid reason.

Q.5. (Total 3 Marks)

The current in an electric motor depends on both the applied electromotive force (e.m.f.) and the back electromotive force (e.m.f.).

Describe the given statement in THREE points.

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Q.6. (Total 2 Marks)

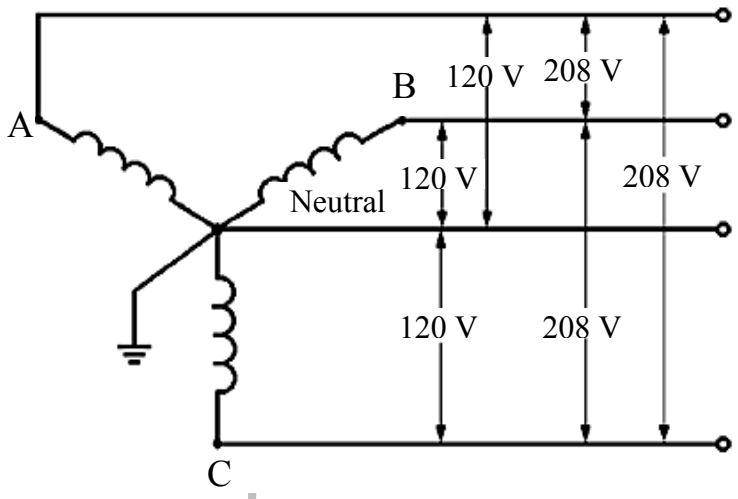
Describe what happens when a radiowave encounters an obstacle such as concrete barriers, rocks, mountains, etc. if the radiowave is modulated via:

Amplitude Modulation (AM).

Frequency Modulation (FM).

Q.7. (Total 3 Marks)

i. Identify the given circuit diagram. (1 Mark)

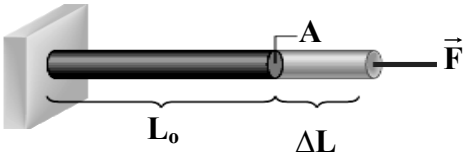


ii. Describe the working of the given circuit. (2 Marks)

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Q.8. (Total 3 Marks)

In the given diagram, $\vec{F}$ denotes the stretching force, **A** is the cross-sectional area, $\mathbf{L_o}$ is the original length of the rod, and $\Delta\mathbf{L}$ is the increase in length.



Derive the formula for the magnitude of the stretching force $\vec{F}$ applied perpendicularly to the surface at the end with the help of Young's modulus.

Q.9. (Total 3 Marks)

Fill in the blanks.

i. For AND gate, output = 1, if all inputs = _____.

ii. For NOR gate, output = _____ if any input = 1.

iii. For _____ gate, output = 0 if all inputs = 1.

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Q.10. (Total 3 Marks)

a. State the Stefan-Boltzmann Law of black body radiations. (1 Mark)

b. Write the mathematical representation of Stefan-Boltzmann Law. (2 Marks)

Q.11. (Total 4 Marks)

a. Does the transition for the emission of characteristic X-rays depend on the nature of the target material atoms? Give reason to support your answer. (2 Marks)

b. Can the X-rays be reflected, refracted, diffracted and polarised just like any other waves? Give reason to support your answer. (2 Marks)

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Q.12. (Total 2 Marks)

State TWO reasons due to which nuclear radiations are potentially harmful to humans.

Q.13. (Total 3 Marks)

a. Does a quark possess a charge magnitude smaller than that of an electron? (1 Mark)

b. According to the quark model of hadrons, how are mesons different from baryons? (2 Marks)

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Q.14. (Total 5 Marks)

EITHER

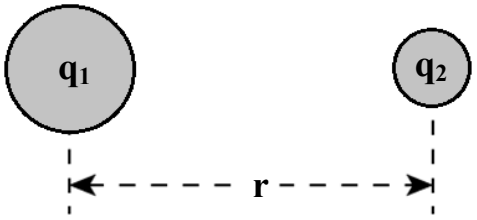
a.

i. Discuss the effect observed after the application of potential difference across the plates of a capacitor. (1 Mark)

ii. Derive the expression for the energy **E** stored in an uncharged capacitor taking voltage at first plate $\mathbf{V_1}$ and at second plate $\mathbf{V_2}$. (4 Marks)

OR

b. State and explain the law which describes the relation of force between two charged particles $\mathbf{q_1}$ and $\mathbf{q_2}$ and write the value of electrostatic constant. (5 Marks)



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Q.15. (Total 5 Marks)

EITHER

a. A transformer works on 220 V and supplies a load with 4 A. If the ratio of the primary and secondary winding of coils is 2:50 respectively, then calculate the (i) primary current and (ii) secondary voltage in the coils.

OR

b. A 0.2 m long inductor has 8000 turns per metre and its cross-sectional area is 5.6×10^{-3} m^2. If 10 A current is passing through the inductor, then calculate the value of energy stored in it.

(**Note**: The value of $\mu_o = 4 \pi \times 10^{-7}$ Wb/A.m.)

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Q.16. (Total 5 Marks)

EITHER

a. Compare mechanisms of pair production and pair annihilation in FIVE points.

OR

b. Describe Planck's assumption about the black body radiation with the help of mathematical relation on energy and momentum.

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